

VELIKO GRADISTE, Serbia

WMO: 132850

Lat: 44.750N

Lon: 21.517E

Elev: 84

StdP: 100.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-13.7	1.2	-8.2	-10.9	1.5	-6.4	11.7	-1.6	10.3	0.2	1.8	110	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	34.4	20.9	32.6	20.7	30.8	20.2	23.1	29.4	22.2	28.9	21.4	28.1	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.1	25.5	20.2	15.1	24.5	19.3	14.2	23.7	68.6	29.3	65.2	28.8	62.5	28.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.5	7.4	-14.8	36.9	3.7	2.3	-17.4	38.5	-19.6	39.8	-21.7	41.1	-24.4	42.8		
(5)			-15.1	24.4	3.6	1.3	-17.7	25.4	-19.7	26.2	-21.8	26.9	-24.3	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	0.5	2.5	7.0	12.6	17.1	20.7	22.7	22.6	17.7	12.3	7.2	1.9		
	DBStd	8.80	4.47	4.84	4.52	3.99	3.57	3.46	3.10	3.19	3.80	4.34	4.61	4.25		
	HDD10.0	1026	293	212	113	21	1	0	0	0	0	27	107	253		
	HDD18.3	2699	551	443	351	174	67	16	3	4	56	190	334	510		
	CDD10.0	1801	0	2	21	99	221	319	393	391	232	99	24	1		
	CDD18.3	434	0	0	0	3	28	86	138	137	38	3	0	0		
	CDH23.3	4632	0	0	1	49	288	924	1511	1456	366	37	0	0		
	CDH26.7	1837	0	0	0	7	56	339	670	653	108	3	0	0		
(14) Wind	WSAvg	2.5	2.8	2.9	2.9	2.9	2.3	1.9	1.8	1.9	2.2	2.7	3.3	2.7		
(15) Precipitation	PrecAvg	692	46	44	42	61	72	79	75	54	64	56	50	51		
	PrecMax	923	86	90	119	113	141	178	199	154	165	142	152	126		
	PrecMin	439	9	3	9	4	20	13	4	6	13	2	0	7		
	PrecStd	136	22	25	26	28	37	43	49	38	36	36	30	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	17.5	22.3	27.5	30.3	34.5	37.2	36.9	32.9	26.8	20.6	13.4		
		MCWB	8.3	10.1	12.6	15.6	18.5	20.9	21.6	21.7	19.4	16.9	13.1	9.0		
	2%	DB	10.1	14.2	19.5	24.4	28.2	32.2	34.4	34.5	29.5	24.1	17.8	10.8		
		MCWB	7.3	8.9	11.0	14.6	18.0	21.1	21.0	20.9	18.4	15.8	12.8	8.2		
	5%	DB	8.3	11.5	16.9	22.1	26.7	30.5	32.5	32.5	27.3	21.8	15.7	9.1		
		MCWB	6.4	7.7	10.1	13.4	17.5	20.5	20.8	20.5	17.7	14.8	11.7	7.1		
	10%	DB	6.4	9.5	14.6	20.0	24.7	28.7	30.5	30.5	25.2	19.5	13.9	7.4		
		MCWB	4.8	6.6	9.0	12.7	16.8	19.9	20.5	20.0	17.1	14.1	10.9	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	11.1	13.2	17.1	20.8	24.2	24.1	24.2	21.3	18.2	14.6	10.0		
		MCDB	11.2	15.2	19.9	24.0	26.8	29.8	29.9	31.0	28.3	23.3	17.9	12.1		
	2%	WB	7.8	9.5	11.8	15.5	19.4	22.7	22.9	22.9	19.8	16.7	13.4	8.5		
		MCDB	9.8	13.2	17.7	22.2	25.4	28.8	29.7	29.1	26.1	21.9	16.6	10.5		
	5%	WB	6.4	8.1	10.7	14.4	18.4	21.7	22.1	21.9	18.7	15.7	12.3	7.2		
		MCDB	8.1	11.0	15.6	20.4	24.2	27.9	29.1	28.7	24.5	20.2	15.2	8.8		
	10%	WB	5.0	6.8	9.6	13.4	17.6	20.7	21.4	21.0	17.9	14.7	11.1	5.9		
		MCDB	6.3	9.4	13.9	18.6	23.1	26.8	28.1	27.9	23.3	18.8	13.7	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.7	10.1	10.9	11.5	12.2	13.0	13.2	11.3	9.9	7.3	5.5		
		MCDBR	8.9	12.6	15.0	15.5	14.7	15.3	16.6	16.7	15.1	13.8	10.3	8.1		
	5% WB	MCWBR	6.2	7.5	7.6	6.7	5.9	5.7	5.4	5.2	5.5	6.2	5.6	5.7		
		MCWBR	8.1	11.3	12.7	12.9	12.9	13.5	13.9	13.6	12.3	12.2	8.9	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.381	0.419	0.480	0.465	0.456	0.464	0.480	0.450	0.407	0.371	0.348		
		taud	2.276	2.175	2.145	2.000	2.104	2.174	2.155	2.104	2.150	2.259	2.308	2.310		
	Ebn at Noon		738	780	803	783	811	819	807	775	763	744	708	700		
		Edh at Noon	85	113	133	167	155	145	146	149	131	103	83	76		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.06	3.28	4.54	5.48	6.11	6.17	5.50	3.84	2.60	1.58	1.03		
	RadStd		0.15	0.27	0.43	0.47	0.55	0.60	0.49	0.54	0.41	0.32	0.16	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.74	N/A	N/A	+0.99	+0.72	+0.89	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.66	N/A	N/A	N/A	+0.36	N/A	N/A	-150	+106	N/A

Nomenclature: See separate page